



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Executive Summary

Goal:

- To predict the likelihood of the Falcon 9's First stage landing successfully
- To gather information about SpaceX and use the data to support the project

Methodologies and approach:

- Gather information via web and API scraping
- Explore the data using SQL and data visualization. Folium used to explore specific requirements
- Finally, use Classification machine learning models to predict the success of the first stage landing

Result

- Decision Trees are best at predicting the success of the first stage landing with a 94% accuracy.
- Launch sites situated closer to the equator are more likely to succeed

Introduction

Background

- The emergence of rising market competitors like Virgin Galactic, Rocket Lab and Blue Origin is happening.
- Successful landing allowed SpaceX to be able to reuse the Falcon 9 and be sold for 62 million dollars compared to other competitors, selling rockets for 165 million dollars
- Space Y, a startup, will aim to compete with SpaceX and will begin by understanding and using data to predict the success of the Falcon 9's first landing.

Section 1

Methodology

Methodology

Executive Summary

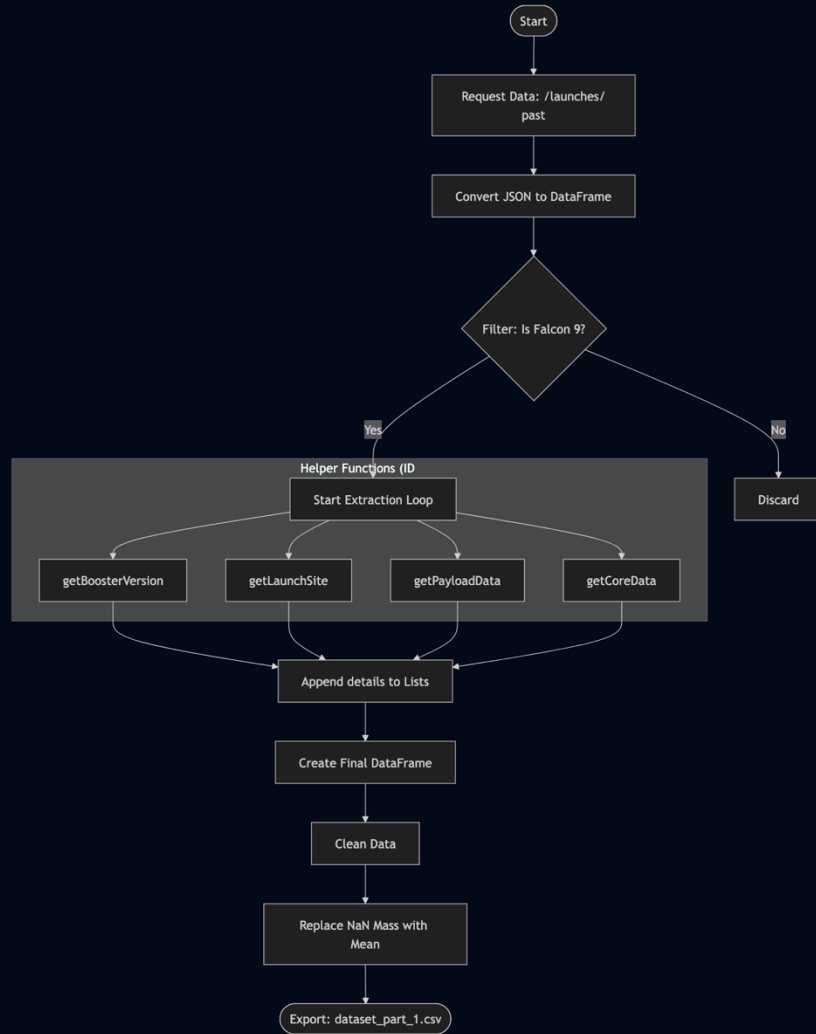
- Data collection methodology:
 - Web scraping from Wikipedia via an API scraping process
- Perform data wrangling
 - Numpy, mean imputation and appending helpful columns.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - Logistic Regression, KNN, SVM and Decision Trees were used and optimized.

Data Collection

- Data was retrieved via Web extraction methods
- Two Web extraction methods were used:
 - API Scraping
 - Web Scraping
- The next two slides contain flowcharts of two different web extraction processes.

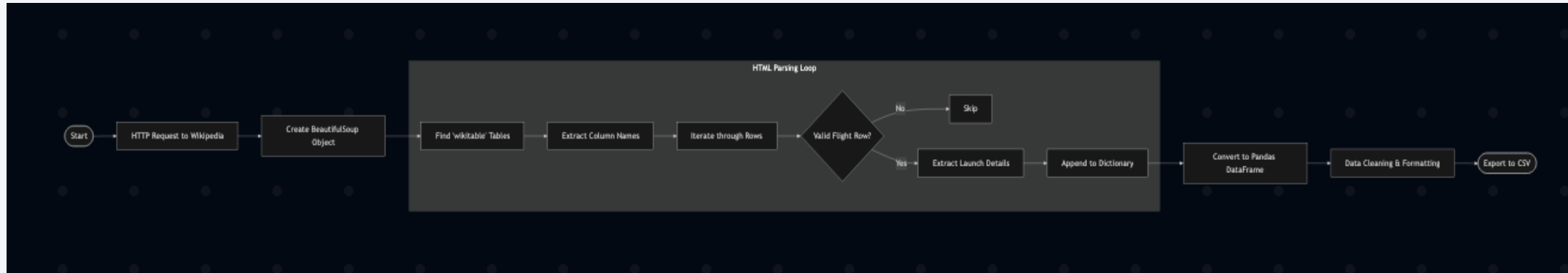
Data Collection – SpaceX API

- Key Takeaways
- Begins by requesting information from the SpaceX API
- Transforms the data obtained from the SpaceX API to make it suitable for analysis and manipulation
- Begins a preparation of the data
 - This leans into wrangling and is why there are two Github URLs referring to this notebook in the next slide



Github URL: <https://github.com/UnhintedSplash/IBM-Data-Science-Capstone-Repository/blob/a306b7e2e98aede327c5800d467db33e1671384e/jupyter-labs-spacex-data-collection-api.ipynb>

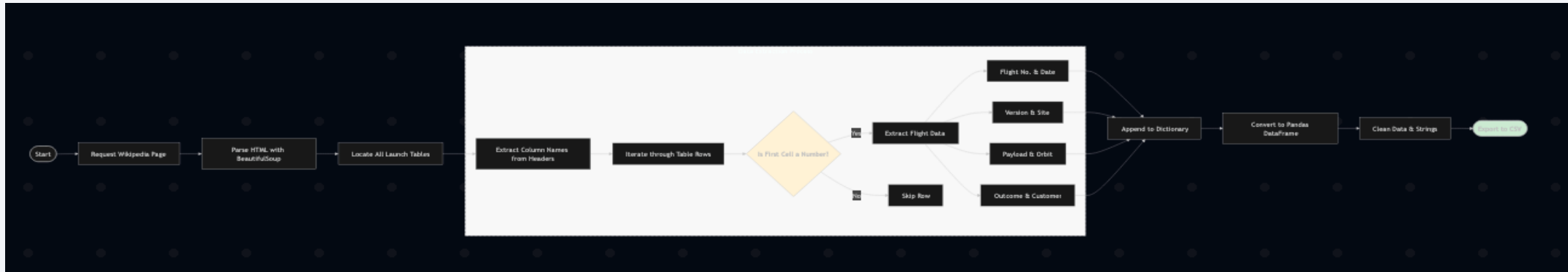
Data Collection - Scraping



Key Takeaways

- Obtain an HTTP request to the source website
- Allow the machine to undergo a loop to process the data
- Turn obtained data into a pandas data frame

Data Wrangling



Key Takeaways

Pandas was used to determine and handle missing values

Mean imputation was used to replace missing values, found only in the Payload Mass column

Github URLs:

- <https://github.com/UnhintedSplash/IBM-Data-Science-Capstone-Repository/blob/a306b7e2e98aede327c5800d467db33e1671384e/jupyter-labs-spacex-data-collection-api.ipynb>
- <https://github.com/UnhintedSplash/IBM-Data-Science-Capstone-Repository/blob/a306b7e2e98aede327c5800d467db33e1671384e/labs-jupyter-spacex-Data%20wrangling.ipynb>

EDA with Data Visualization

The following plots were used for said purposes:

- Scatter Plot – To determine relationships between different variables
- Bar Chart – To compare measures for different categories
- Line Chart– To help determine trends over a certain period

EDA with SQL

- The SQL queries were used to obtain:
 - Unique launch site names
 - Launch sites beginning with 'CCA'
 - Total and average Payload Mass
 - First ever successful landings
 - Filtered datasets
 - Total number of successful and failed landings
 - Boosters with maximum payload mass
 - Records from 2015
 - Ranking landing outcomes by frequency

Build an Interactive Map with Folium

Objects used and purpose:

- Circles – To mark the locations of different launch sites
- Marked Clusters – To group up launch outcomes found in the same location
- Lines – To determine the distance from a launch site's closest coastline to the closest city

Build a Dashboard with Plotly Dash

- Plots:
 - Pie Charts – to view and compare success to failure ratios for each launch site
 - Scatter plots – to help understand the relationship between payload mass and launch outcomes
- Interactions:
 - Launch Site options – to help users navigate through different launch sites or a thorough comparison of all launch sites
 - Range Slider – to help look through a range of payload masses and comparing against launch outcomes

Predictive Analysis (Classification)



How it was built – Using Scikit Learn for tools that help make and improve models



For a reliable score – Using Standardization, Grid Search Cross Validation and train test splits



To evaluate model performance – A confusion matrix is produced which allows for accuracy, precision and F1 score calculations



The best performing model after evaluation – Decision Tree



The next slide presents the flowchart of the process

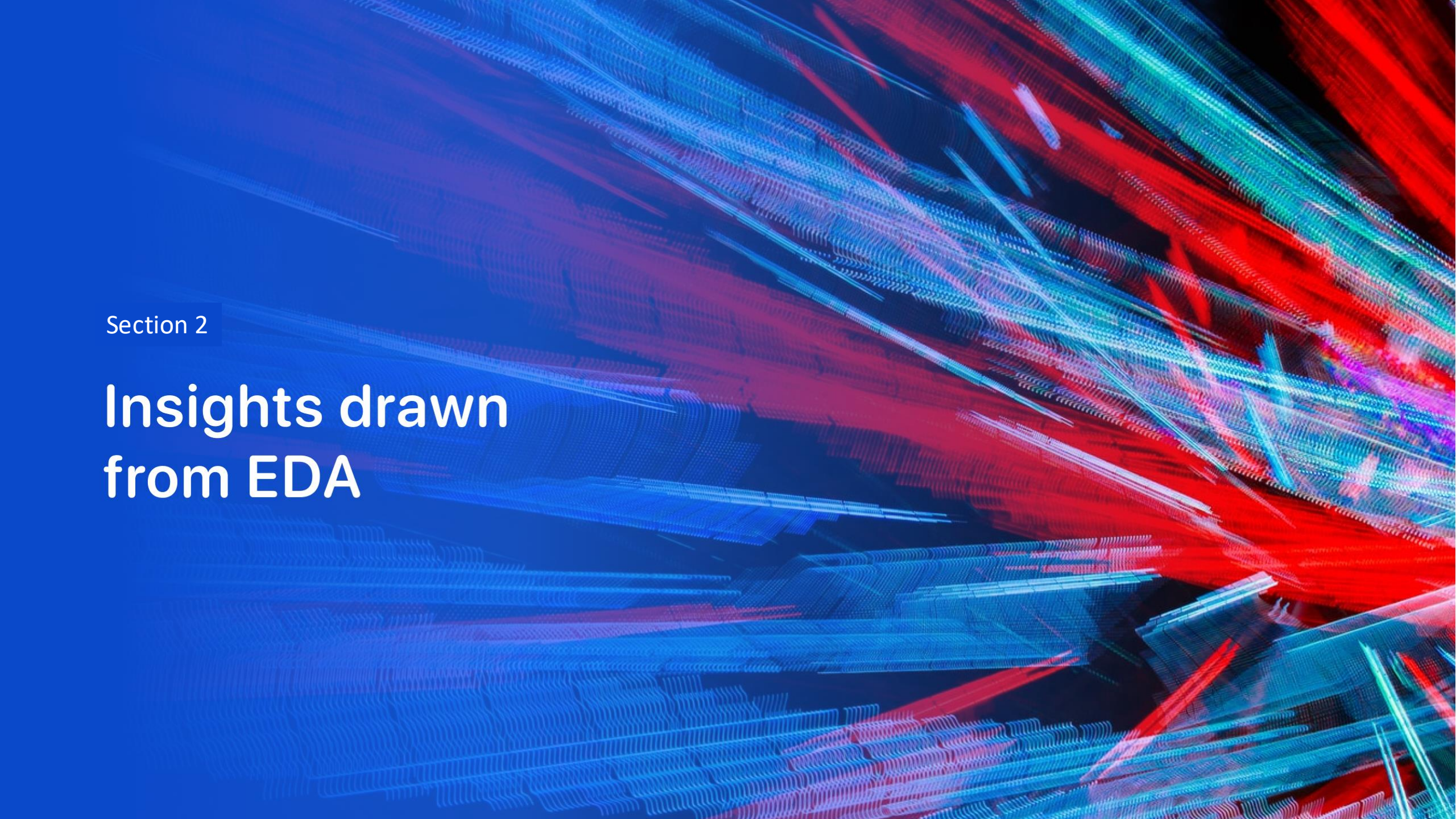
Predictive Analysis Flow Chart(Classification)



Results

Contents of results explored:

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

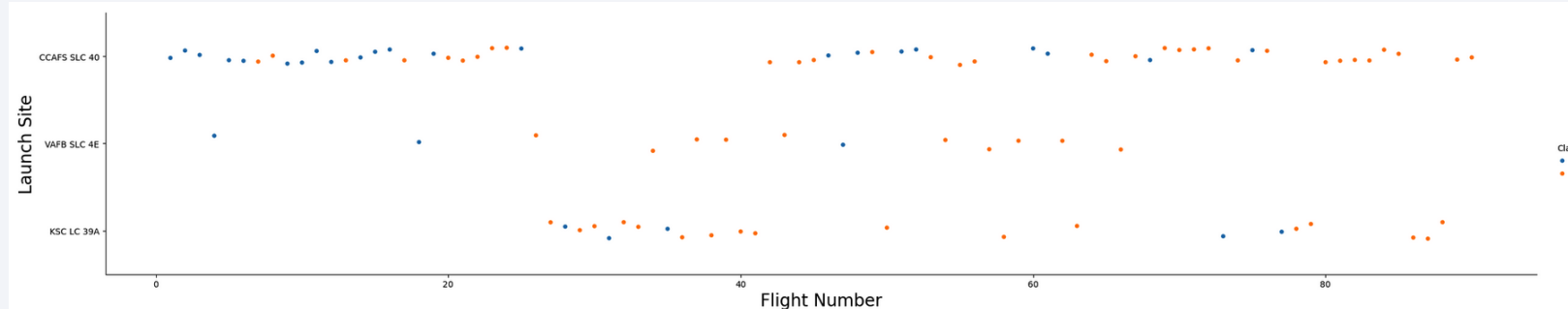
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

The scatter point chart below shows:

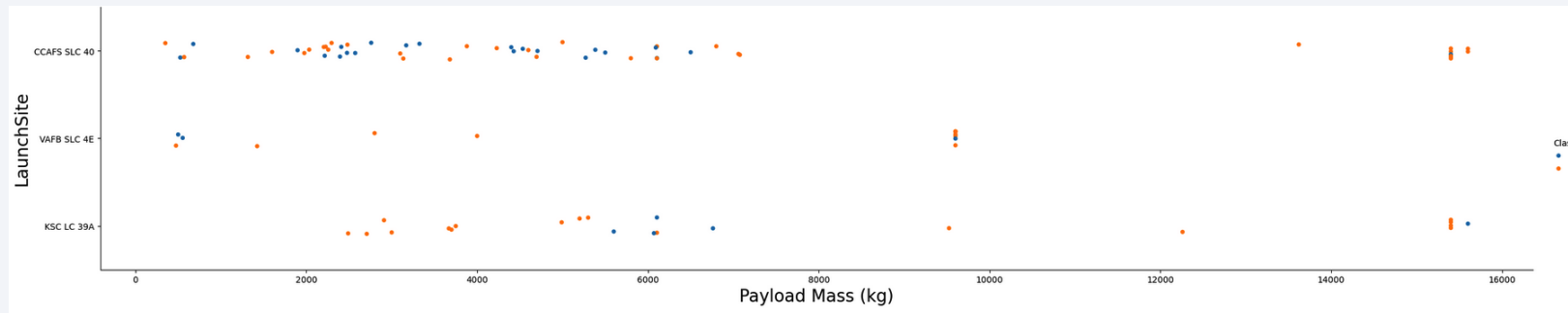
- Earlier launches tended to fail whilst later launches tended to succeed
 - Development is impacting the rocket's landing outcome successfully and early failures are worth documenting
- Flights are frequently documented on the “CCAFS SLC 40” launch site
 - Information about landing for this launch site is abundant – helpful for future studies



Payload vs. Launch Site

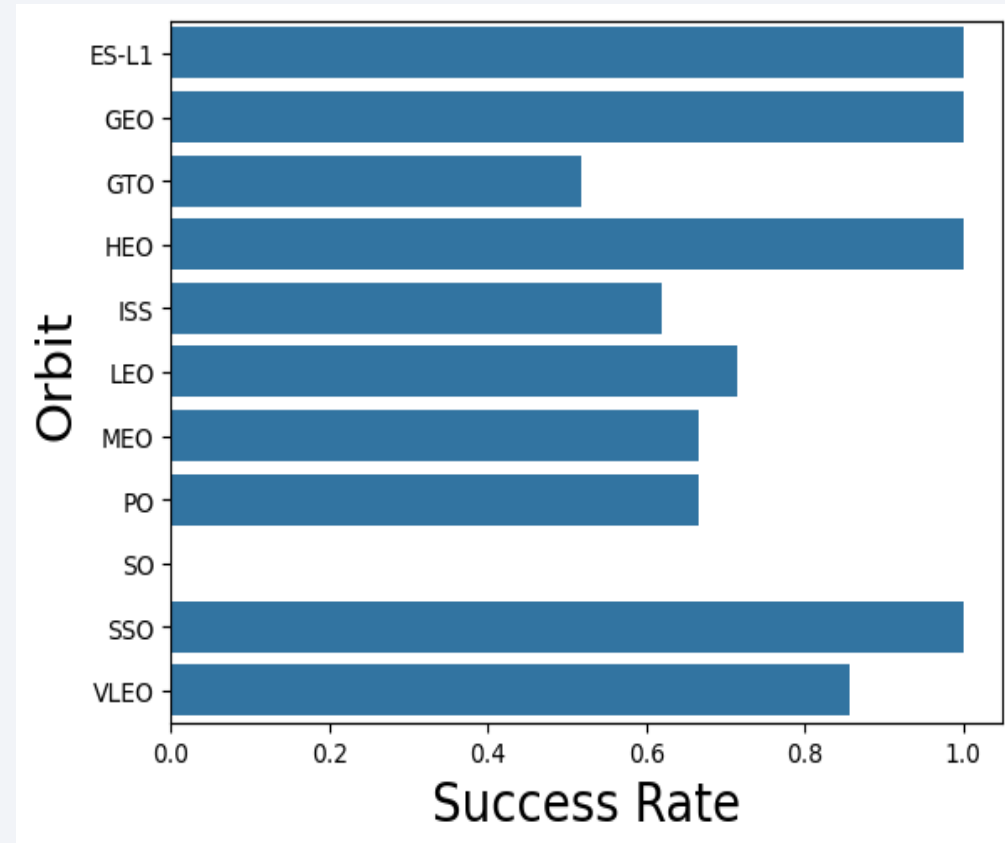
The scatter point chart below shows:

- Payload is dense for each rocket in each launch site is generally below 8000
 - Information is abundant for lower payloads
- When scaled, higher payload masses(above 8000) tend to be more successful than lower payload masses (below 8000)
 - More information will have to be gained about higher mass rockets before confirmation



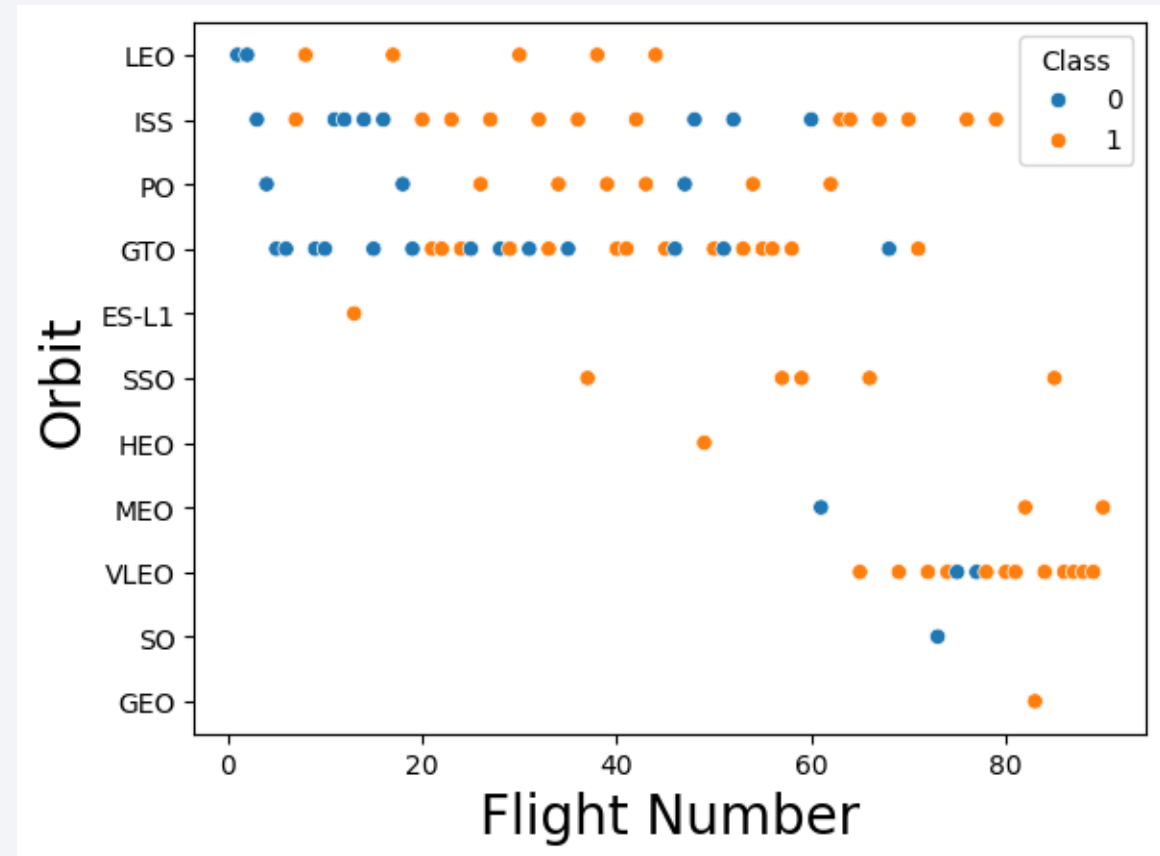
Success Rate vs. Orbit Type

- Certain and impossible success rates lack sufficient information
 - We cannot consider the data reliable now(Refer to the Appendix)
- GTO and ISS contain more data
 - It would be appropriate to use these orbits as a basis for determining landing outcomes



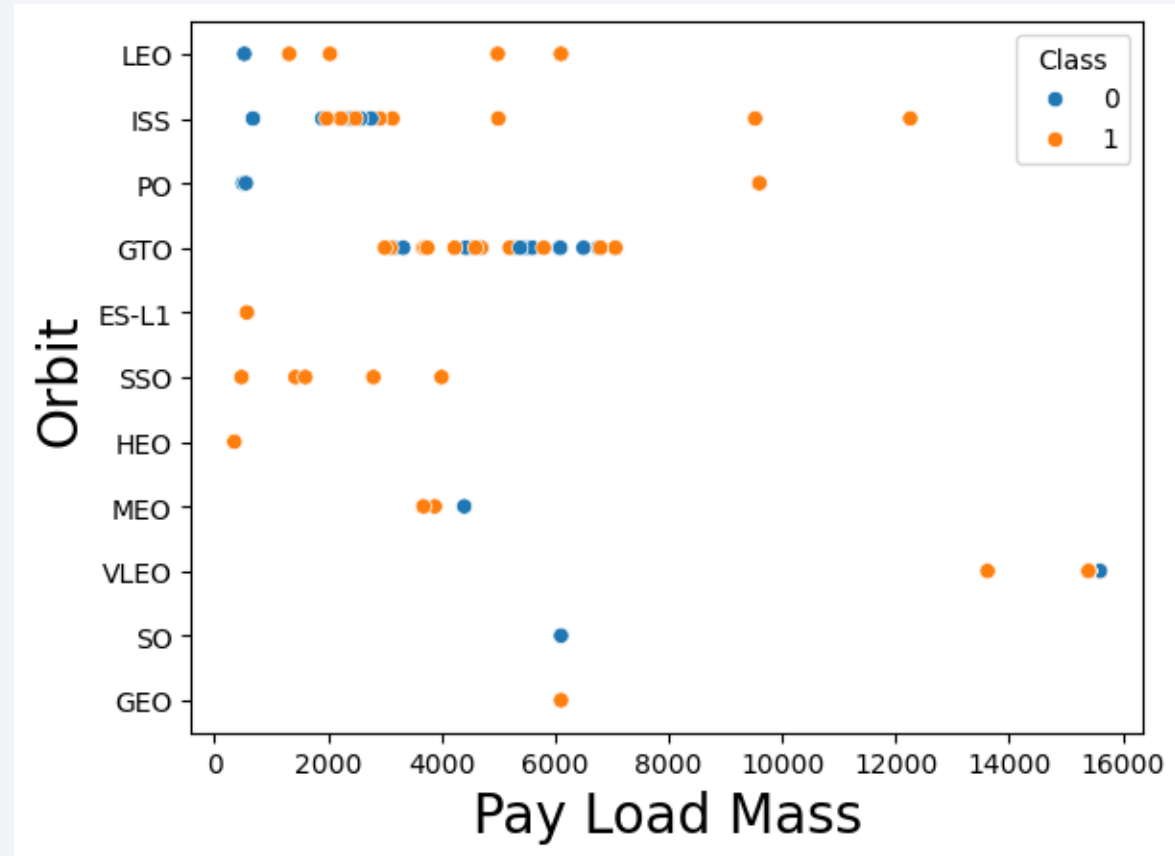
Flight Number vs. Orbit Type

- Later and more successful flights tend to occur in the VLEO orbit
 - Further studies will need to be conducted to determine if orbit transition is a leading cause of higher success rate
 - The ISS orbit trials will supply information to support this study due to being chronologically unbiased.



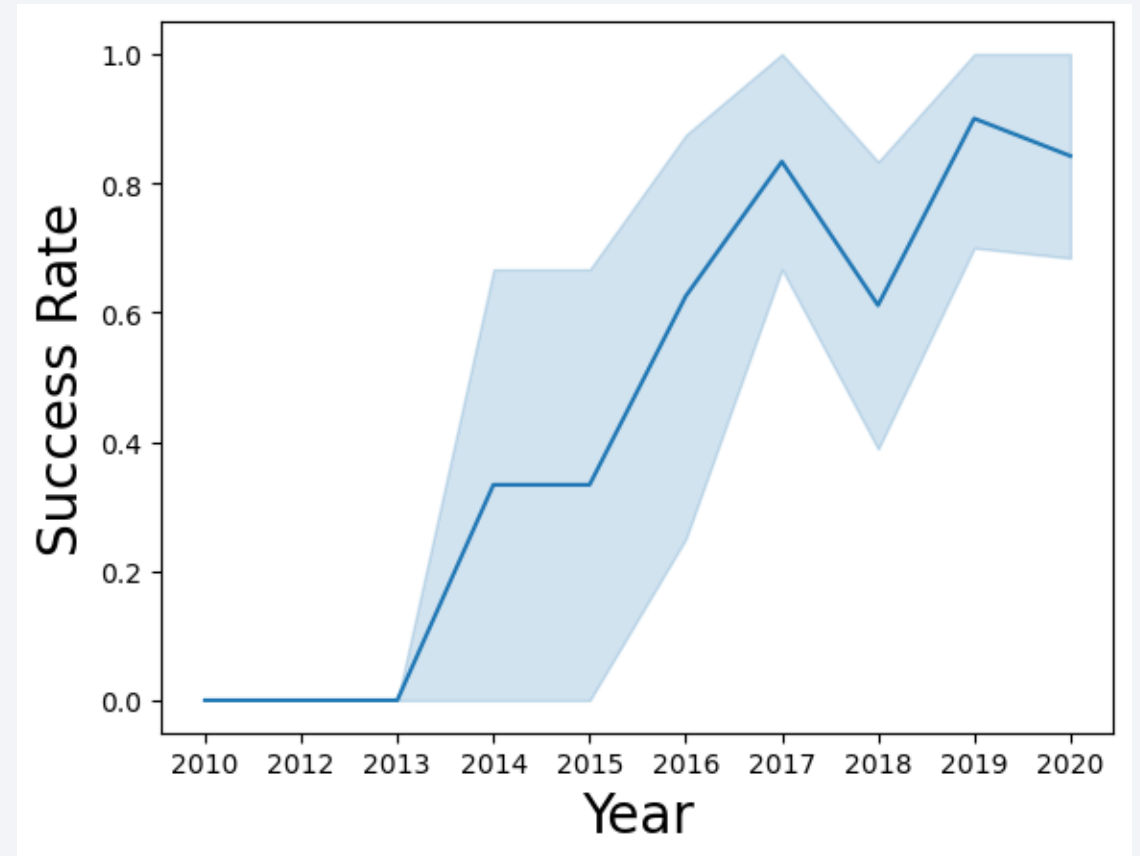
Payload vs. Orbit Type

- GTO and ISS have a densely packed mass range
 - Rockets launched into the GTO and ISS orbit will have an abundance in information regarding their respective mass range and landing outcomes
- Heavier masses (above 8000) tend to be launched into the VLEO and ISS orbit where it often tends to succeed



Launch Success Yearly Trend

- The trend corresponds to the earlier 'Flight number vs Launch Site' scatter point chart.
 - By reliable means, it is confirmed that development has led to an increase in success rate overall
 - Development in rocket builds are seeing a positive trend.



All Launch Site Names

Below shows the different launch sites we will be referring to for comparison purposes and insights extraction

```
[11]:
```

```
query1 = 'SELECT DISTINCT "Launch_Site" FROM SPACEXTABLE'  
result1 = pd.read_sql_query(query1, con)  
print(result1)
```

```
   Launch_Site  
0  CCAFS LC-40  
1  VAFB SLC-4E  
2    KSC LC-39A  
3  CCAFS SLC-40
```

Launch Site Names Begin with 'CCA'

- We may use the 5 site names starting with 'CCA' for minor information helpful for data analysis.

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
[12]: query2 = 'SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE "CCA%" LIMIT 5'
      result2 = pd.read_sql_query(query2, con)
      result2.head(6)
```

[12]:	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The total payload mass is calculated below for report findings or future calculations.

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[13]: query3 = 'SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE'  
      result3 = pd.read_sql_query(query3, con)  
      print(result3)
```

```
      SUM("PAYLOAD_MASS_KG_")  
0                619967
```

Average Payload Mass by F9 v1.1

- Below is the average payload mass by a specific booster version: F9 v1.1, for insights into mass units utilized for a specific booster version.

Task 4

Display average payload mass carried by booster version F9 v1.1

```
[39]: query4 = 'SELECT AVG("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "BOOSTER_VERSION" LIKE "F9 v1.1%"'
      result4 = pd.read_sql_query(query4, con)
      print(result4)

      AVG("PAYLOAD_MASS_KG_")
0                2534.666667
```

First Successful Ground Landing Date

- The first successful landing outcome found to present within documents or reports as a possible referral.

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
[43]: query5 = 'SELECT MIN("Date") FROM SPACETABLE WHERE "Mission_Outcome"="Success"'
      result5 = pd.read_sql_query(query5, con)
      print(result5)

      MIN("Date")
0    2010-06-04
```

Successful Drone Ship Landing with Payload between 4000 and 6000

- The query result below shows a set of booster versions which succeeded a drone ship landing with payload masses between 4000 and 6000.
- This should be helpful insight into the structure of specific builds.

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[45]: query6 = 'SELECT "BOOSTER_VERSION" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG" BETWEEN 4000 AND 6000'
      result6 = pd.read_sql_query(query6, con)
      print(result6)
```

```
Booster_Version
0      F9 v1.1
1    F9 v1.1 B1011
2    F9 v1.1 B1014
3    F9 v1.1 B1016
4      F9 FT B1020
5      F9 FT B1022
6      F9 FT B1026
7      F9 FT B1030
8    F9 FT B1021.2
9    F9 FT B1032.1
10   F9 B4 B1040.1
11   F9 FT B1031.2
12   F9 B4 B1043.1
13   F9 FT B1032.2
14   F9 B4 B1040.2
15   F9 B5 B1046.2
16   F9 B5 B1047.2
17   F9 B5 B1046.3
18     F9 B5B1054
19   F9 B5 B1048.3
20   F9 B5 B1051.2
21     F9 B5B1060.1
22   F9 B5 B1058.2
23     F9 B5B1062.1
```

Total Number of Successful and Failure Mission Outcomes

- Below is the total number of successful and failure mission outcomes.
- A total of 100 successful landings alongside only a single failed mission.
- This will be helpful for reports, documentation and manufacturing improvements

▼ Task 7 ¶

List the total number of successful and failure mission outcomes

```
[63]: query7 = '''SELECT "Mission_Outcome", COUNT(*) AS count FROM SPACEXTABLE
WHERE "Mission_Outcome"~
IN ('Success', 'Failure (in flight)', 'Success (payload status unclear)', 'Success ')
GROUP BY "Mission_Outcome"'''
result7 = pd.read_sql_query(query7, con)
print(result7)
```

	Mission_Outcome	count
0	Failure (in flight)	1
1	Success	98
2	Success	1
3	Success (payload status unclear)	1

Boosters Carried Maximum Payload

- We find that 11 booster versions carry the maximum payload mass
- Simple tweaking can be made to determine the landing outcomes; however, it will help inform the making of future booster versions.

```
[14]: query8 = '''SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG"=(SELECT MAX("PAYLOAD_MASS_KG") FROM SPACEXTABLE)'''
      result8 = pd.read_sql_query(query8, con)
      print(result8)
```

	Booster_Version
0	F9 B5 B1048.4
1	F9 B5 B1049.4
2	F9 B5 B1051.3
3	F9 B5 B1056.4
4	F9 B5 B1048.5
5	F9 B5 B1051.4
6	F9 B5 B1049.5
7	F9 B5 B1060.2
8	F9 B5 B1058.3
9	F9 B5 B1051.6
10	F9 B5 B1060.3
11	F9 B5 B1049.7

2015 Launch Records

- During 2015, we find that only 2 rockets failed landing, most commonly to be aiming on a drone ship.
- Studying their failures may help lead to a full set of successful landing outcomes for an entire year.

```
query9 = '''SELECT substr("Date", 6, 2) AS month, "Landing_Outcome", "Booster_Version", "Launch_Site" FROM SPACEXTABLE
WHERE substr("Date", 0, 5)="2015" AND "Landing_Outcome" LIKE "Failure%"""
result9 = pd.read_sql_query(query9, con)
print(result9)
```

	month	Landing_Outcome	Booster_Version	Launch_Site
0	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
1	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Within the range of the dates: 2010-06-04 and 2017-03-20, we find that the the most frequent landing outcome was a category where no landing was attempted
- This is then followed by drone ship landings which are equal regarding failure and successful proportions.

```
• [26]: query10 = '''SELECT "Landing_Outcome", COUNT(*) AS COUNT FROM SPACEXTABLE
WHERE "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome" ORDER BY count DESC''' ~
result10 = pd.read_sql_query(query10, con)
print(result10)
```

	Landing_Outcome	COUNT
0	No attempt	10
1	Success (drone ship)	5
2	Failure (drone ship)	5
3	Success (ground pad)	3
4	Controlled (ocean)	3
5	Uncontrolled (ocean)	2
6	Failure (parachute)	2
7	Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

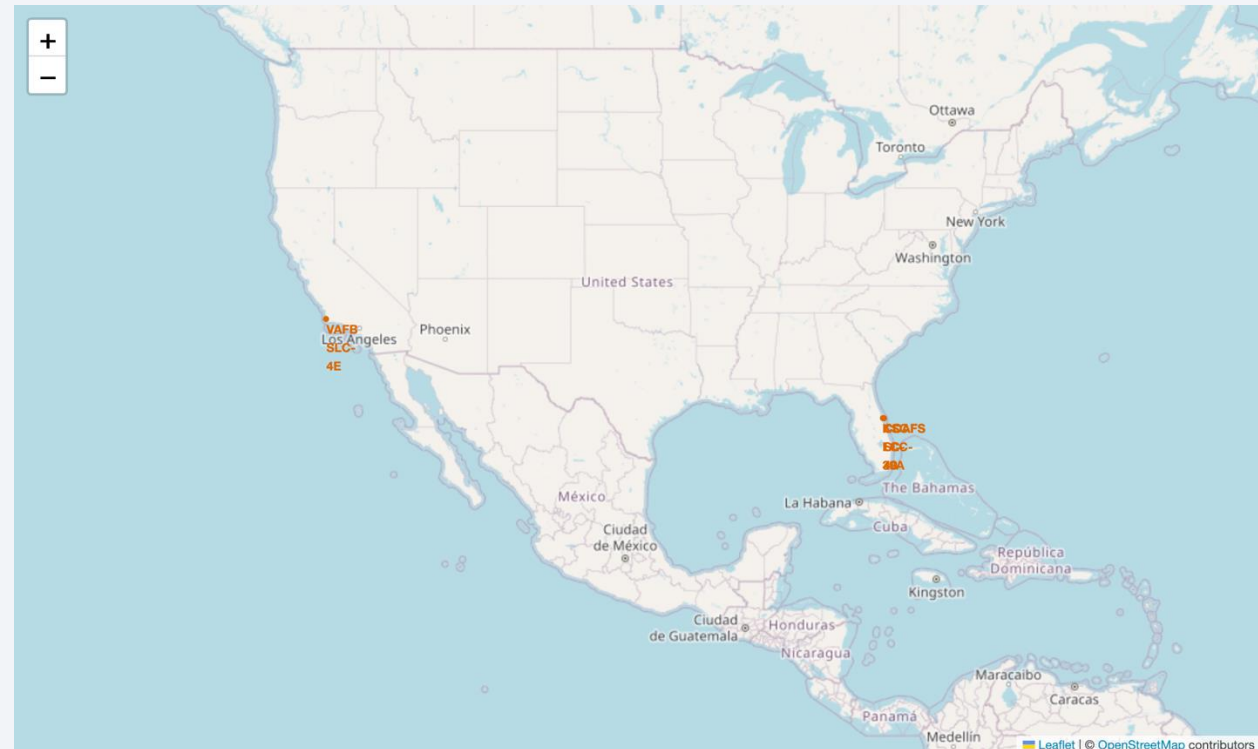
Section 3

Launch Sites Proximities Analysis

Location of Launch Sites

On the map we find that:

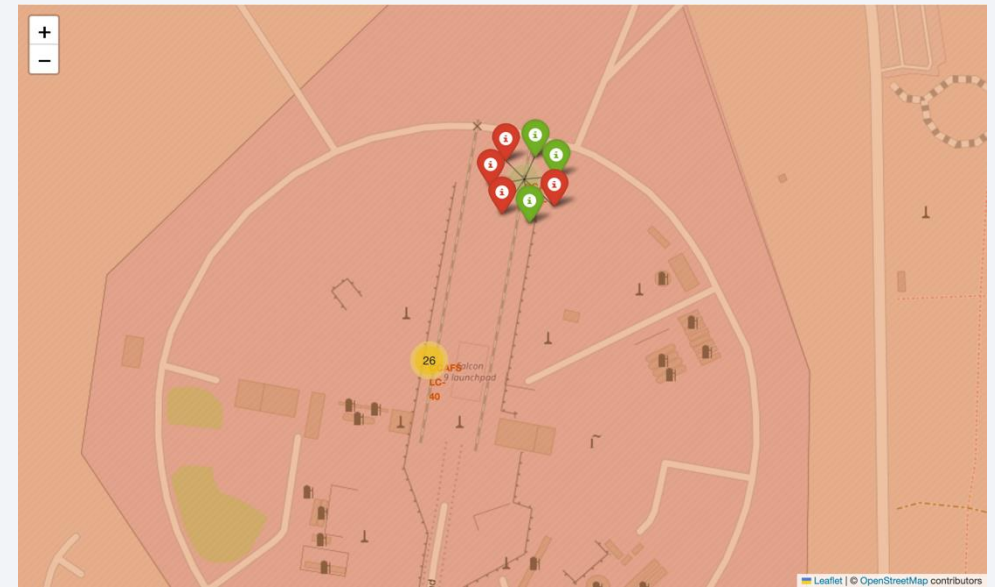
- Launch Sites are located near the coastline
 - Launching is following a safe protocol where rockets will not interfere with the public space if it fails
- Launch sites are close to the equator
 - Allows rockets to reach orbit fast and thus less resource is consumed for trials



Clusters of launch outcomes

The map of the 'CCAFS LC-40' outlines success markers (green) and failure markers (red).

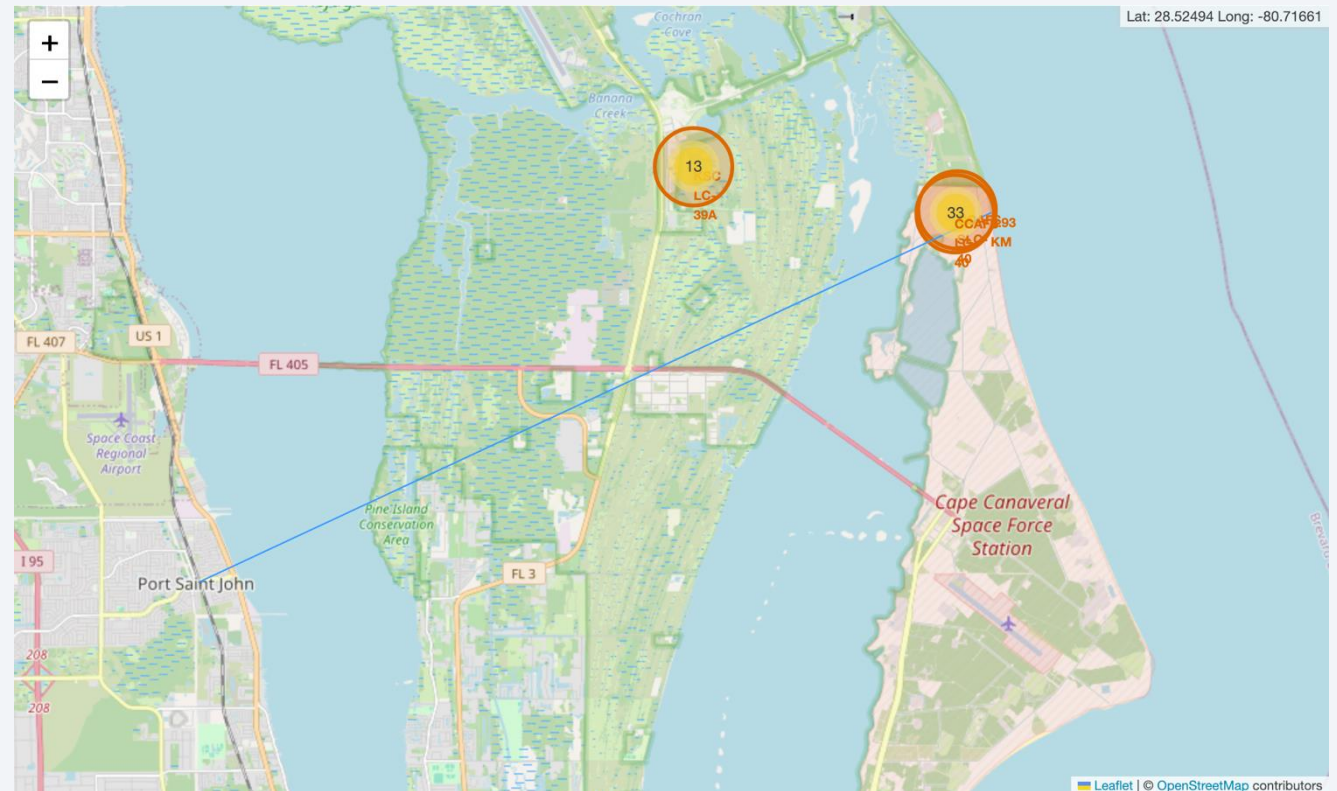
- Getting to take a look into a launch site as such informs of the success rate
 - This will help us compare location success rates and determine the greatest location for launching and landing rockets

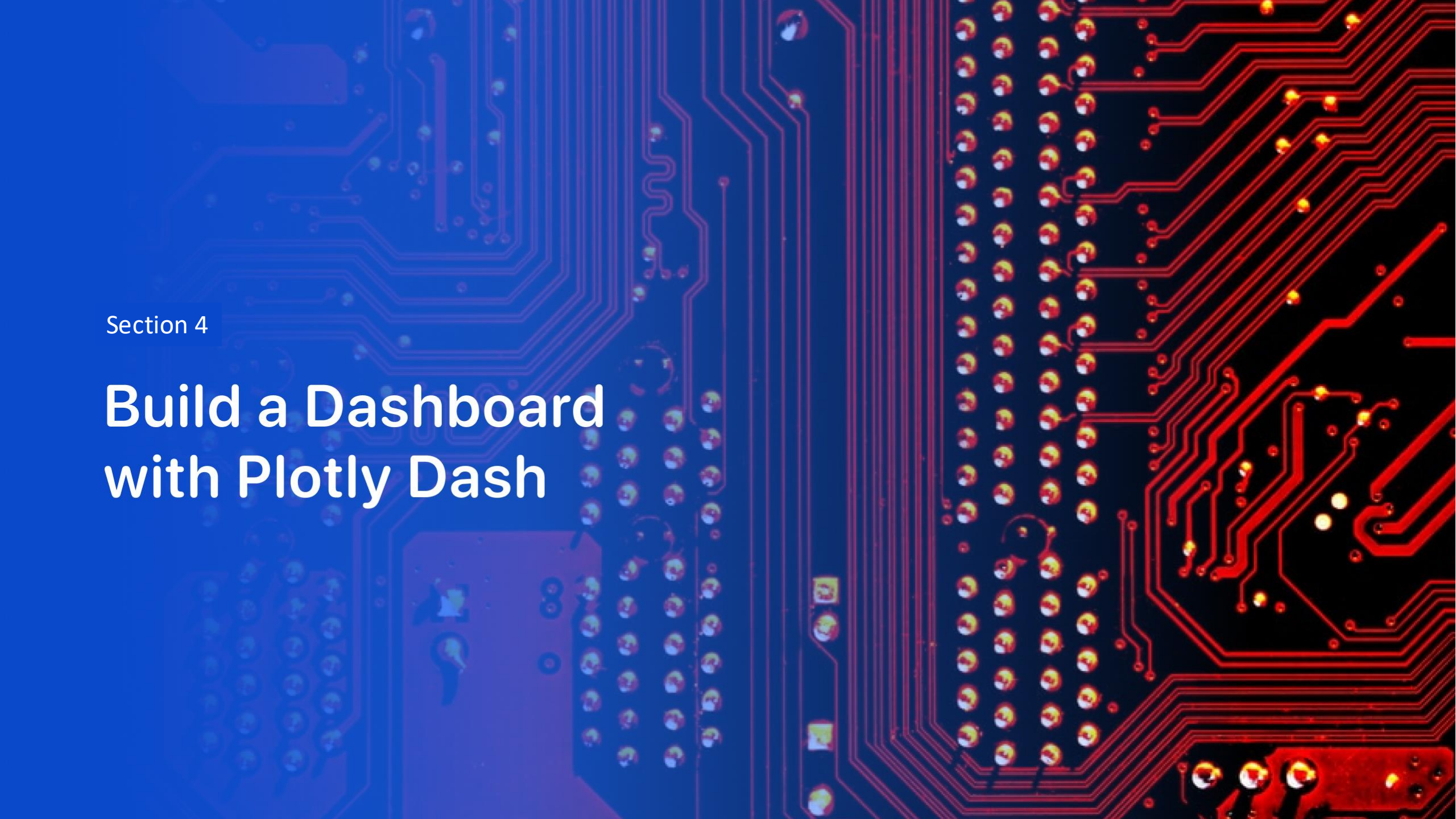


Distance from coastline to the nearest city

The nearest observed city is **Port Saint John**, where the distance from the city to the coast line is 22.73km

- If a failure were to occur past the coastline, the nearest city will not be in radius of being affected by a lethal accident
- However, if it were to happen behind the coastline, infrastructure will be affected by a lethal accident, though the city would be barely affected.
- Future studies should support this claim.



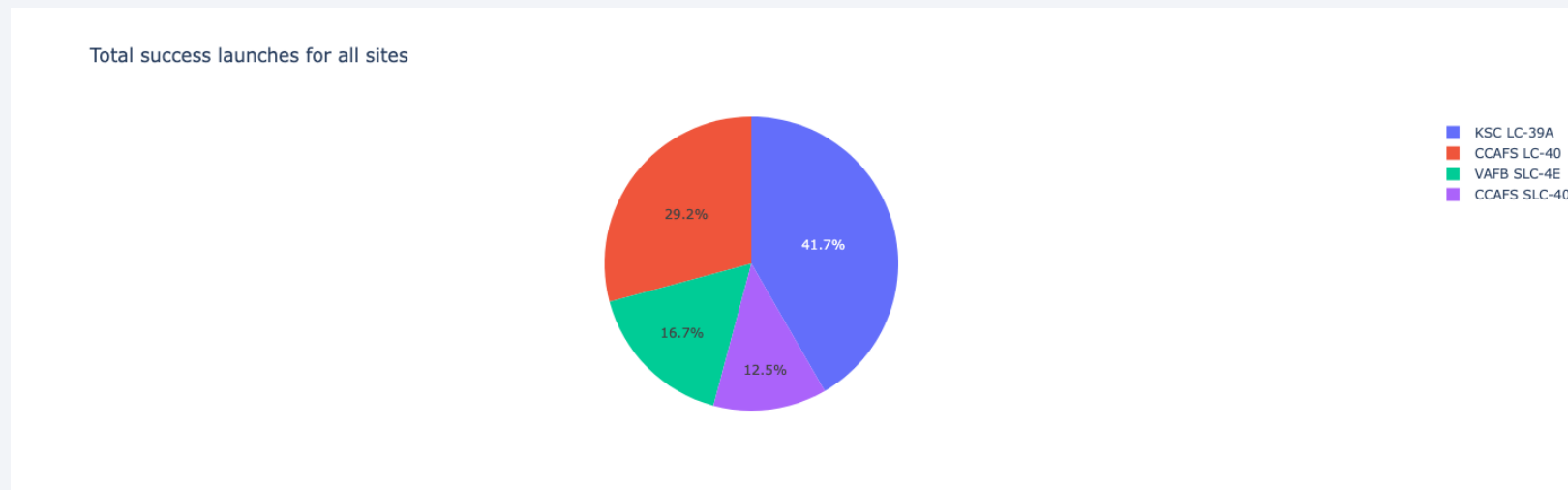


Section 4

Build a Dashboard with Plotly Dash

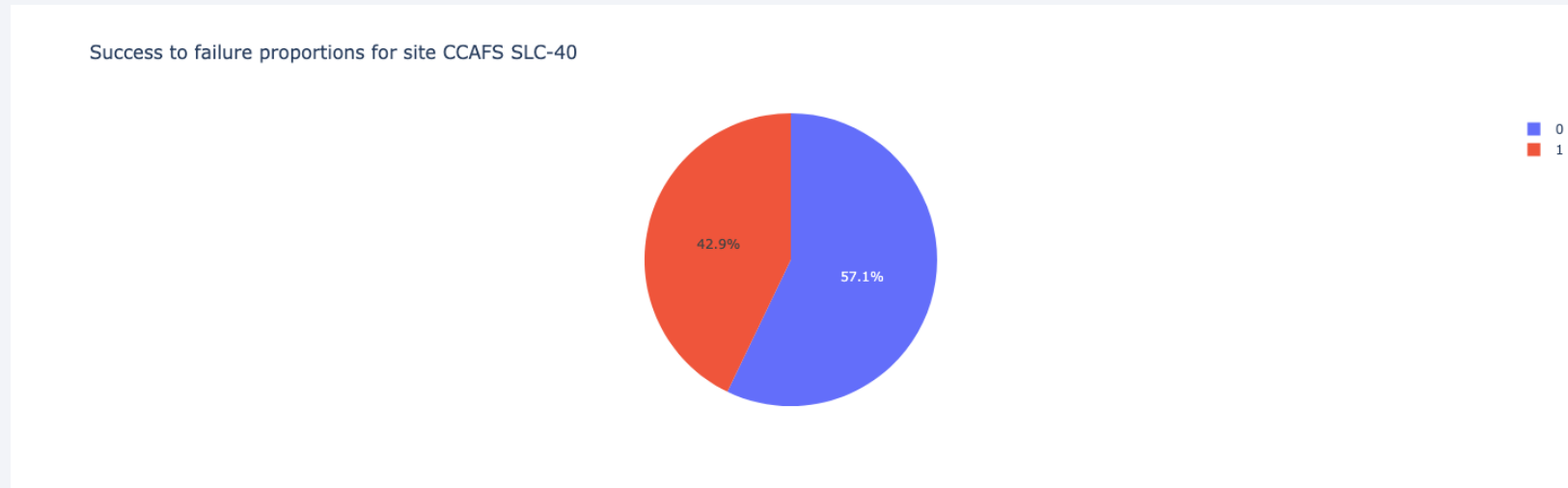
Pie Chart of successful landings for all sites

- The pie chart below shows that KSC LC-39A is responsible for majority of successful landings
 - It infers that it may be reasonable to do launches from KSC LC-39A due to this finding
- CCAFS sites also perform well
 - Looking back into the map, we find that all these launch sites (CCAFS and KSC) are located towards eastern America
 - Geographically, it is more reasonable to do landings situated towards the eastern area of the Americas



Pie chart of the site with the highest success ratio

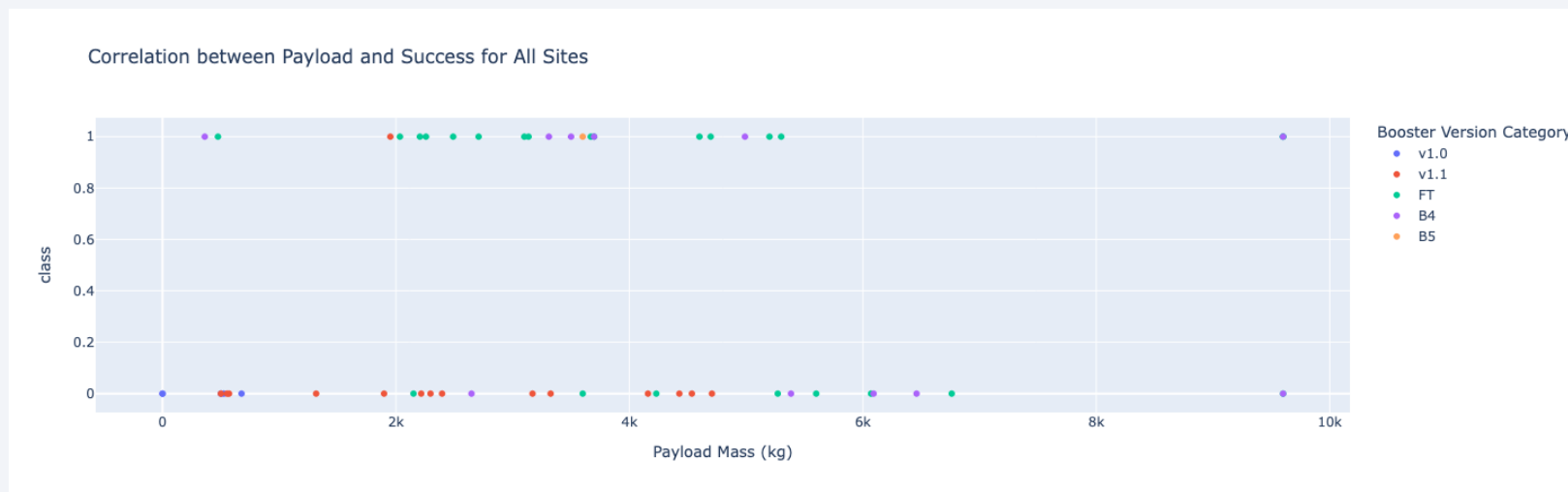
- The pie chart below shows the launch site with the highest success to failure ratio
 - This launch site is located towards the eastern area to further imply that a higher success rate can be obtained by setting launch site locations to the east



Launch outcome against Payload mass for all sites

Below shows a binary scatterplot chart indicating each Booster version under all sites

- We find that more failures were made than success for all ranges of mass
- Overall, the FT Booster version often succeeds and v1.1 booster often fails
 - This can bring further insight into future Booster version considerations, whether it be updating booster versions or producing newer models





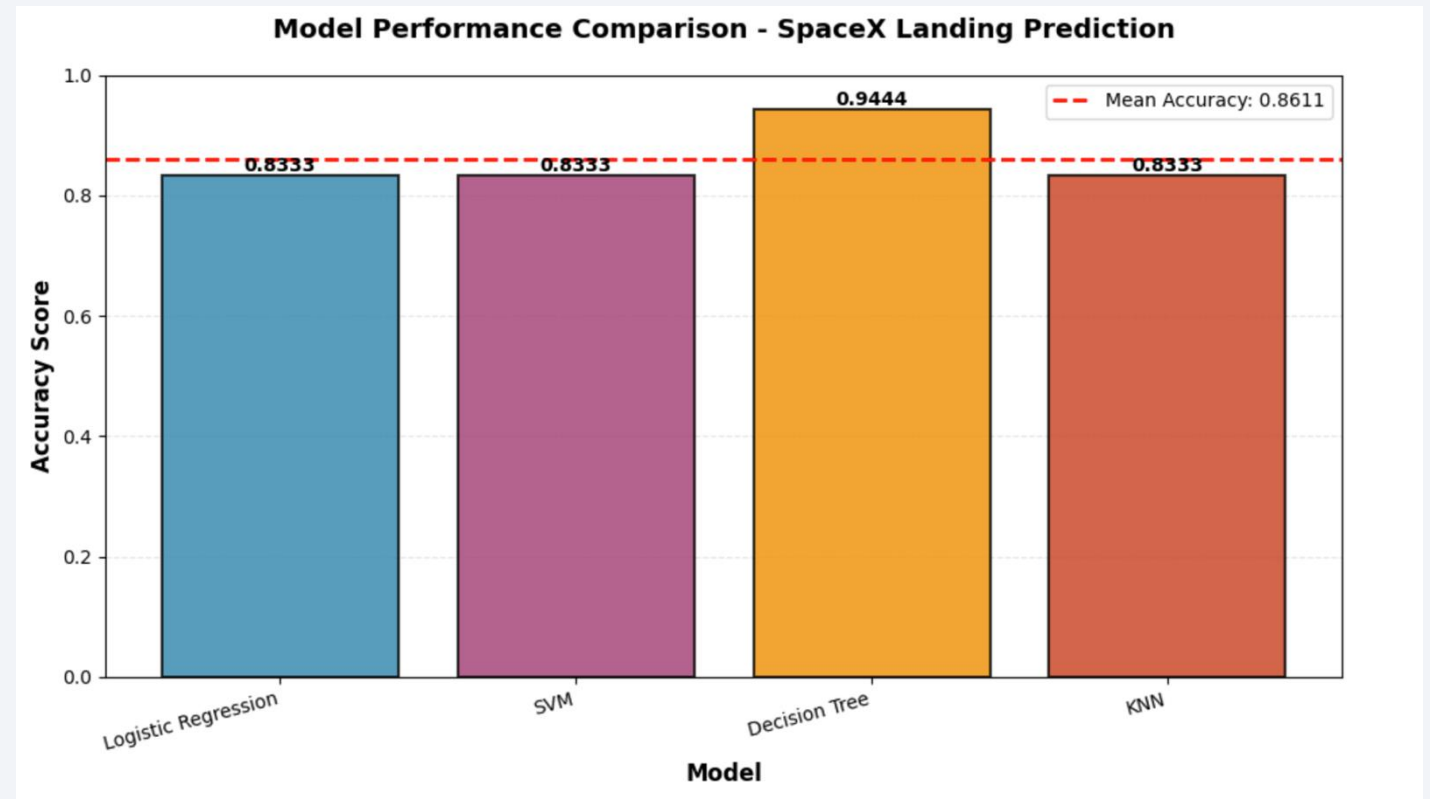
Section 5

Predictive Analysis (Classification)

Classification Accuracy

The bar graph to the side shows that the best performing classification model is the **Decision Tree**.

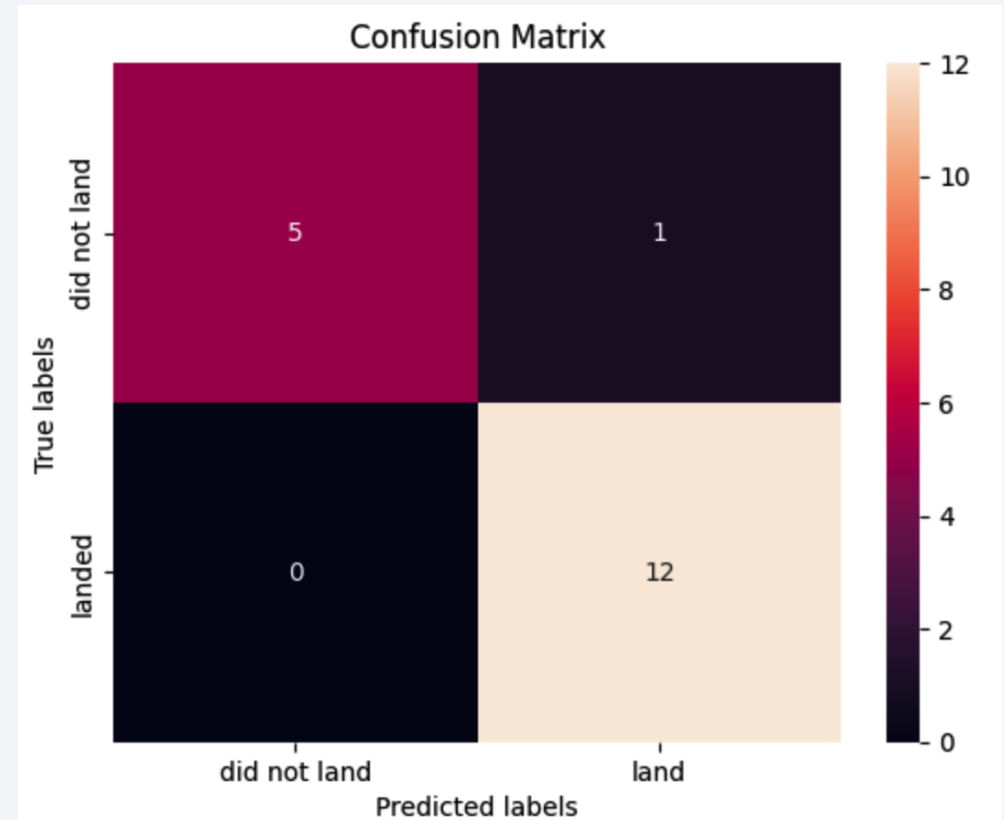
The main measure of comparison is model accuracy.



Confusion Matrix

This is the confusion matrix for the decision tree in which there only exists one false positive and zero false negatives.

Using this confusion matrix, we find that decision trees will be reliable at predicting landing outcomes.



Conclusion and key takeaways

- With visual data exploration, it was determined that **GTO and ISS orbits, and lighter payloads are reliable variables to study.**
- With SQL data exploration, **important details were extracted** which are key to reports, documentation and further assessment.
- Folium maps inform of **geographic awareness and safety** when it comes to rocket launches, enabling an overview about optimal launch locations.
- **Dashboard features** and graphs helped determine that the eastern American launch site locations were optimal places for rocket launches.
- A **Decision Tree** should be used to anticipate the outcome of Falcon 9's first landing - it is accurate and reliable.

Appendix

```
In [26]: SaintJohn_lat = 28.4762
SaintJohn_lon = -80.77839

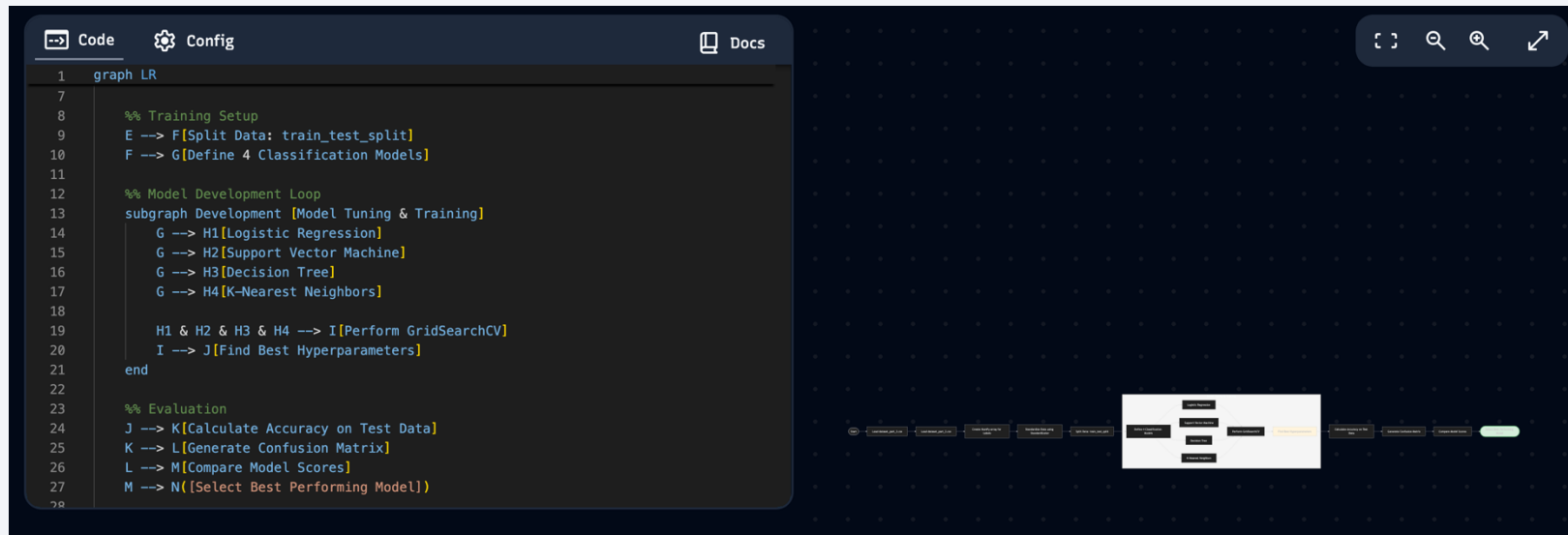
point1 = (SaintJohn_lat, SaintJohn_lon)
point2 = (coastline_lat, coastline_lon)

distance = geodesic(point1, point2).kilometers # or .miles, .meters, etc.
print(f"Distance: {distance:.2f} km")
```

Distance: 22.73 km

A code snippet for calculating the distance from
Port Saint John to the coastline

Appendix



Using mermaid to produce flowcharts
(to underline how great of a tool this is)

Appendix

```
Orbit
GTO      27
ISS      21
VLEO     14
PO        9
LEO        7
SSO        5
MEO        3
ES-L1     1
HEO        1
SO         1
GEO        1
Name: count, dtype: int64
```

These are the orbit counts, and this is just here to prove that some of these orbits cannot be reliable despite consisting of a 1.0 or 0.0 success rate.

Thank you!

